PROJECT PROPOSAL

Project Title: Design and Development of a Web-Based Multi-Organization Lost and Found Management System with Automated Item Matching, Ownership Verification, and Return Process Tracking

University: Astana IT University

Program: Software Engineering

Team Members:

- Makhanbetiyar Begina Berikbolkyzy (SE-2427)
- Amralina Kamila (SE-2427)

Supervisor: Alua Bauyrzhanovna Myrzakerimova

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1. Problem Statement

Losing personal belongings in universities, shopping centers, offices, and other public places is a common problem. Although many organizations have their own lost and found procedures, information about lost and found items is often managed separately through reception desks, security departments, spreadsheets, or informal communication channels.

This fragmented approach makes it difficult for individuals to search for their belongings across different organizations. Users may need to contact several locations individually, while employees may struggle to organize item records, identify potential owners, and track the return process.

Another challenge is verifying ownership before returning an item. Without a structured verification process, organizations may face difficulties determining whether a claimant is the legitimate owner.

Therefore, there is a need for a centralized web-based system that supports multiple organizations while providing structured item registration, automated matching, ownership verification, and return tracking.

2. Project Relevance

The proposed project addresses the need to improve the accessibility, organization, and transparency of lost and found services.

A multi-organization platform could allow users to search for lost belongings through a single interface rather than contacting individual organizations separately. At the same time, participating organizations could manage their own item records and return procedures within a shared system.

Automated item matching could help identify potential connections between lost and found reports based on item attributes such as category, color, location, and date.

The project is relevant to software engineering because it involves requirements analysis, web application development, database design, role-based access control, information matching, security considerations, and usability evaluation.

The practical relevance of the proposed system will be further investigated through interviews with potential users and organization representatives.

3. Target Users

The system is intended for four main user groups.

General Users: Individuals who have lost or found personal belongings and want to submit reports, search for potential matches, or track ownership claims.

Organization Employees: Authorized staff members who register found items, review reports, process ownership claims, and update return statuses.

Organization Administrators: Representatives responsible for managing their organization's employees and overseeing lost and found operations.

Platform Administrator: A system-level administrator responsible for managing participating organizations, platform access, and overall system administration.

4. Project Aim

The aim of this project is to design and develop a web-based multi-organization lost and found management system that supports item registration, automated matching of lost and found reports, ownership verification, and return process tracking.

5. Project Objectives

The main objectives are:

1. To analyze the problems and needs of potential users and organizations involved in lost and found processes.
2. To review relevant academic literature and compare existing lost and found solutions.
3. To define functional and non-functional requirements for the proposed system.
4. To design the system architecture, database structure, and user interfaces.
5. To implement a web-based platform supporting multiple organizations and user roles.
6. To develop an automated mechanism for identifying potential matches between lost and found item reports.
7. To implement an ownership claim and verification workflow.
8. To provide return status tracking and relevant user notifications.
9. To evaluate the system through functional testing and usability assessment.

6. Proposed Solution

The proposed solution is a centralized web platform that connects users and participating organizations through a shared lost and found management environment.

Users will be able to create lost item reports containing descriptive information, including item category, color, approximate location, date, and additional identifying details.

Authorized organization employees will be able to register found items and manage records associated with their organizations.

The system will compare lost and found reports using selected item attributes and identify potential matches. Initially, this functionality is planned as a rule-based or similarity-scoring approach rather than a complex machine learning model.

When a possible match is identified, the user will be able to submit an ownership claim. Authorized employees will review the claim using appropriate verification information before approving the return.

The platform will also provide status tracking so users and employees can follow the progress of a reported item or ownership claim.

7. Key Features

The planned system will include:

- User registration and authentication.
- Role-based access for users, employees, and administrators.
- Registration of lost and found item reports.
- Search and filtering of item records.
- Automated identification of potential item matches.
- Ownership claim submission and verification.
- Return process status tracking.
- Notifications about relevant matches and status changes.
- Organization-specific dashboards.
- Basic administrative management and activity history.

8. Project Scope and Limitations

In Scope

The project will focus on the design and implementation of a responsive web application, including its user interface, backend services, database, and core lost and found workflows.

The platform will support multiple participating organizations while maintaining appropriate access restrictions between organizational records.

Automated matching will initially rely on structured item information and defined comparison rules.

Ownership verification will be supported through a claim review process involving authorized organization employees.

Out of Scope

The initial project does not include:

- Development of a separate native mobile application.
- Physical collection, storage, or delivery of lost items by the platform.
- Integration with every public or private organization in Kazakhstan.
- Guaranteed identification of item ownership without human verification.
- Advanced image recognition or large-scale machine learning models.
- Integration with national identity verification systems.

These capabilities may be considered for future development if appropriate resources and requirements become available.

9. Expected Outcomes

The expected outcome is a functional web-based prototype of a multi-organization lost and found management platform.

The planned deliverables include:

1. Project research and requirements documentation.
2. System architecture and design diagrams.
3. An interactive user interface prototype.
4. A structured database and backend implementation.
5. User and organization management functionality.
6. Lost and found report management.
7. Automated item matching functionality.
8. Ownership verification and return tracking workflows.
9. A verification and evaluation report.

The completed system is expected to demonstrate how a shared digital platform can support more organized and accessible lost and found operations.

10. Project Success Criteria

The project will be considered successful if the implemented prototype satisfies its core functional requirements and demonstrates the main user workflows.

Success will be evaluated based on the following criteria:

- Users can create and manage lost item reports.
- Authorized employees can register and manage found items.
- The system can identify potential matches based on predefined attributes.
- Ownership claims can be submitted, reviewed, and processed.
- Users can view the current status of their reports and claims.
- Organization-specific access restrictions work as intended.
- The main workflows pass defined functional tests.
- Users can complete representative tasks during usability evaluation.

These criteria will be refined during requirements analysis and used to prepare the verification and evaluation plan.

11. Conclusion

The proposed project aims to address the fragmented management of lost and found belongings by developing a shared web-based platform for users and multiple organizations.

By combining structured item registration, automated matching, ownership verification, and return tracking, the system is intended to improve the organization and accessibility of lost and found processes.

The project will apply software engineering principles throughout requirements analysis, system design, implementation, and evaluation, with an emphasis on developing a feasible and testable solution.